Service Bulletin

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EGR Cooler Reuse Guidelines	

EGR Cooler Reuse Guidelines

Product Affected

- All exhaust gas recirculation (EGR) equipped engines.

This document describes specific visual criteria to be used for inspection of exhaust gas recirculation (EGR) coolers.

The new criteria will be used to more accurately diagnose coolant leaks in EGR coolers and to reduce the number of “No Trouble Found” (NTF) coolers returned to Cummins Inc. The previous EGR cooler “Leak Test” improvement made to repair Procedure (011-019 - EGR Cooler) in Section 11, significantly reduced the number of NTF complaints; however, the NTF rate is still at an unacceptable level and necessitated further modification of the procedures.

Eighty percent of the EGR coolers that have coolant loss into the exhaust stream show signs of uneven soot buildup or, in extreme cases, significant buildup of soot in the exhaust outlet.

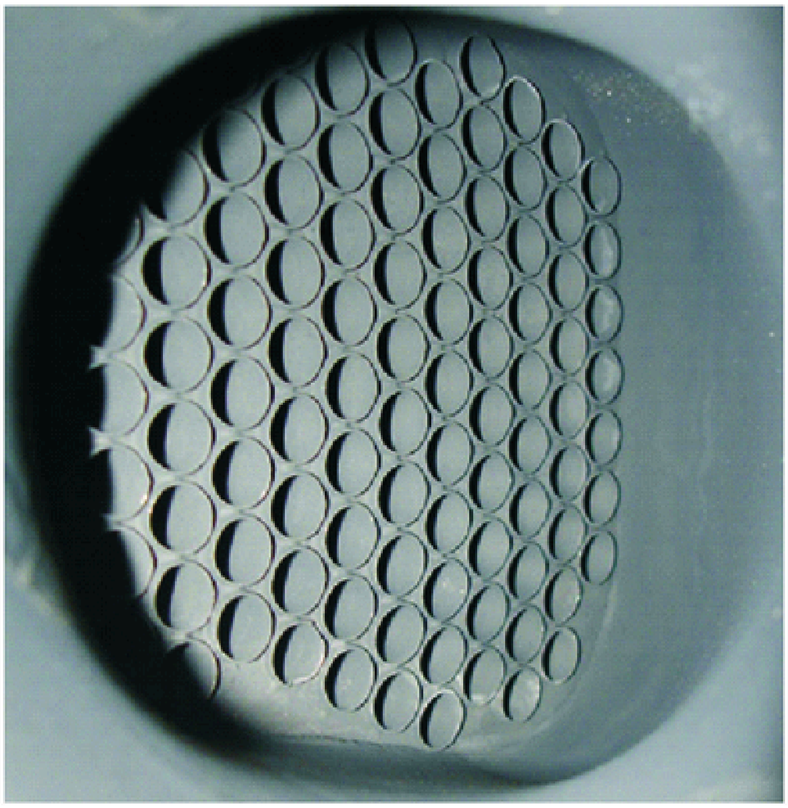
The technician's ability to diagnose leaking EGR coolers can be greatly improved by inspecting the exhaust outlet using this new visual criteria, and by closely following the pressure test in Procedure 011-019 (EGR Cooler) in Section 11, on coolers that show no signs of soot buildup, but are still suspected to be leaking coolant.

Note : This visual inspection is intended to be used in conjunction with the present EGR Cooler Testing procedure and not in place of it.

Visual inspection of the EGR cooler exhaust outlets, as shown in the following photographs, should be used in conjunction with the reuse information in Table 1.

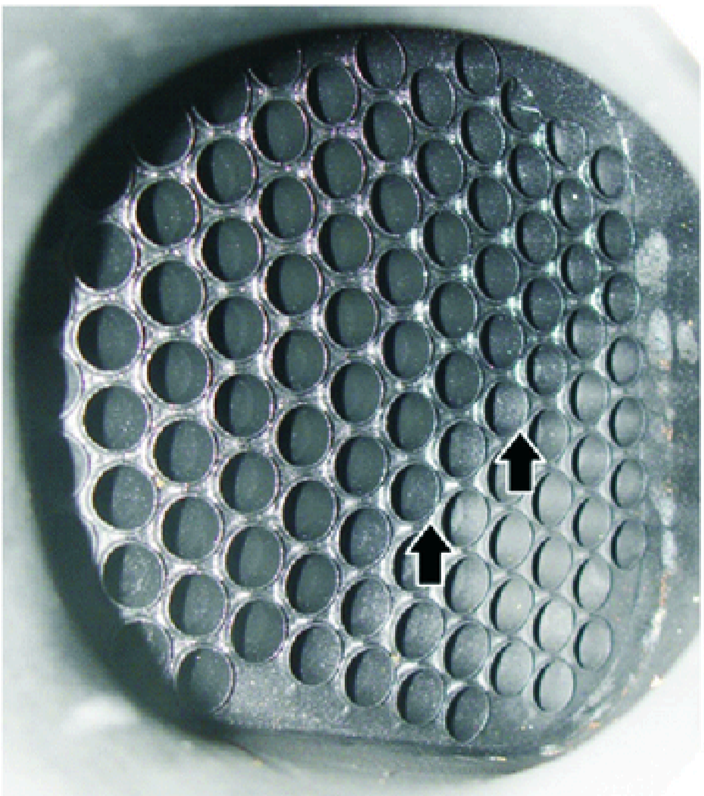
Reference Figures A through J.

Note : A wet condition of the EGR cooler exhaust outlet shall not be used as the only criteria for determining that the EGR cooler is not reuseable.



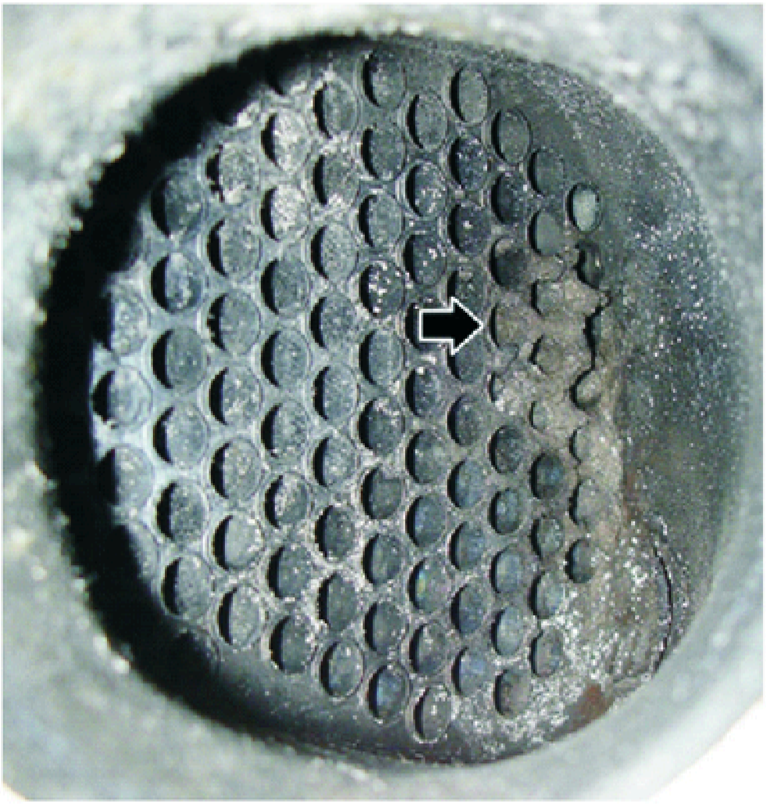
11o00026

Figure A. Even, light soot in the exhaust outlet of the EGR cooler.



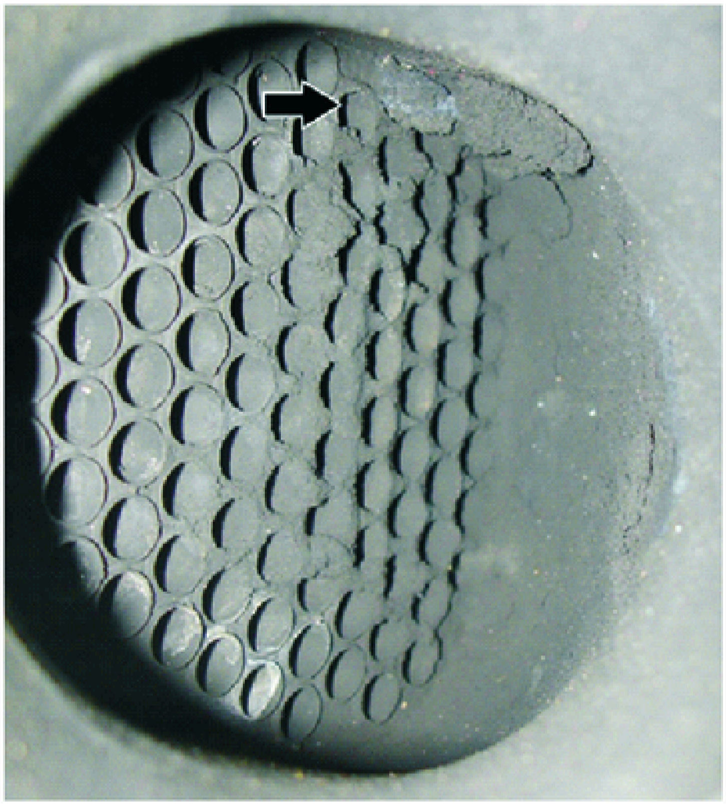
11o00027

Figure B. Even, light wet soot (shown in the upper right half of this photograph).



11o00028

Figure C. Uneven soot buildup, white coolant tracks.



11o00029

Figure D. Uneven soot buildup. Top of the EGR cooler exhaust outlet.

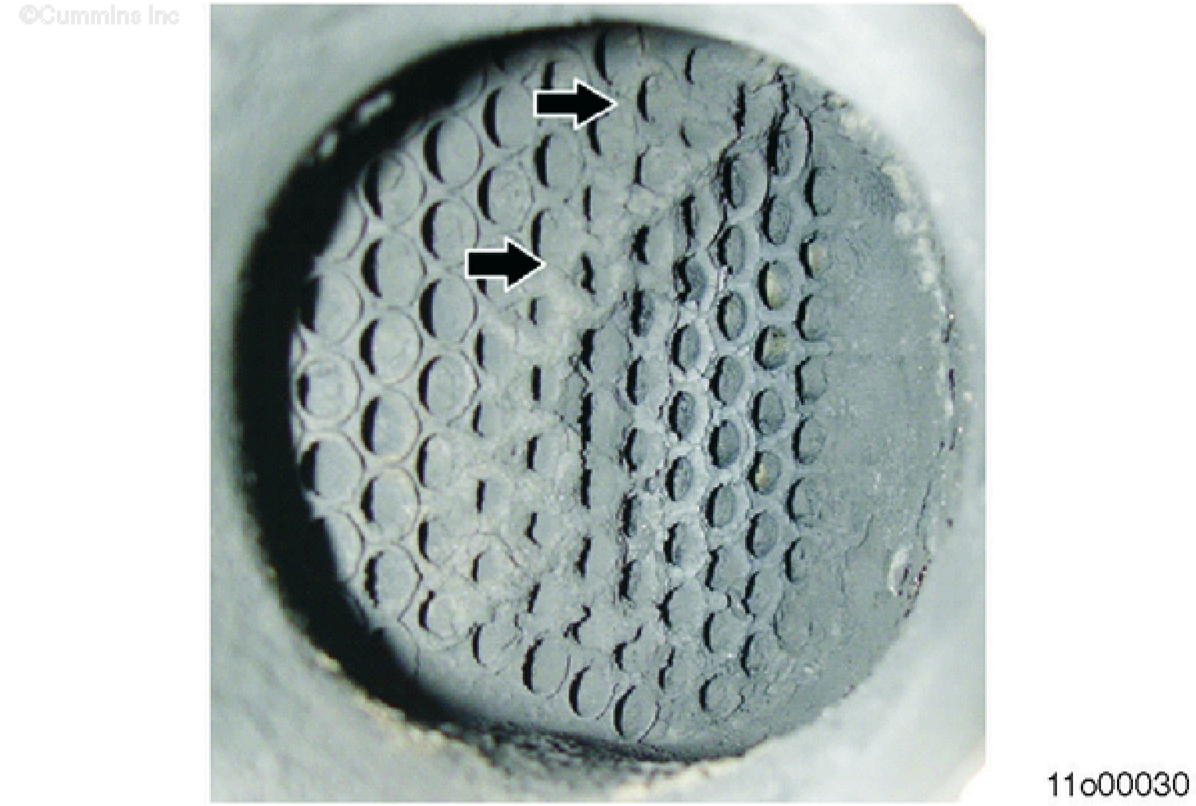


Figure E. Uneven soot buildup. Streaking down the EGR cooler bulkhead in the exhaust outlet.

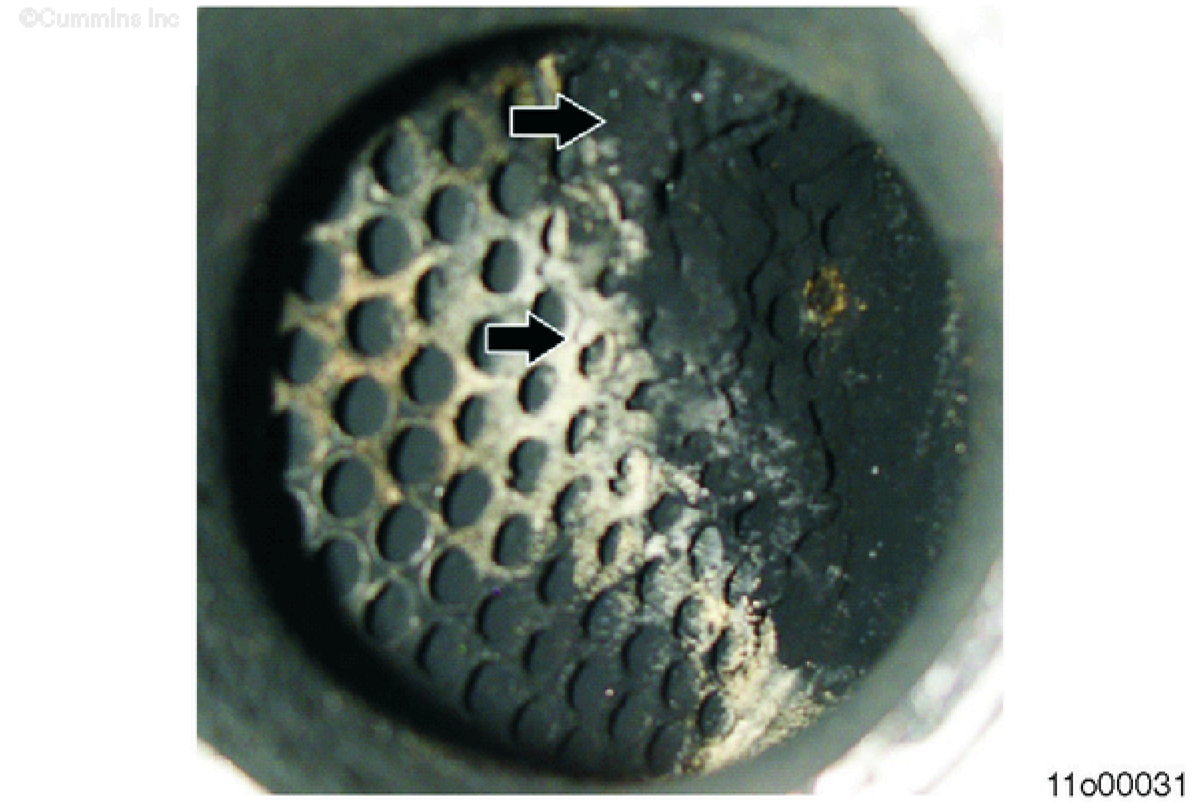


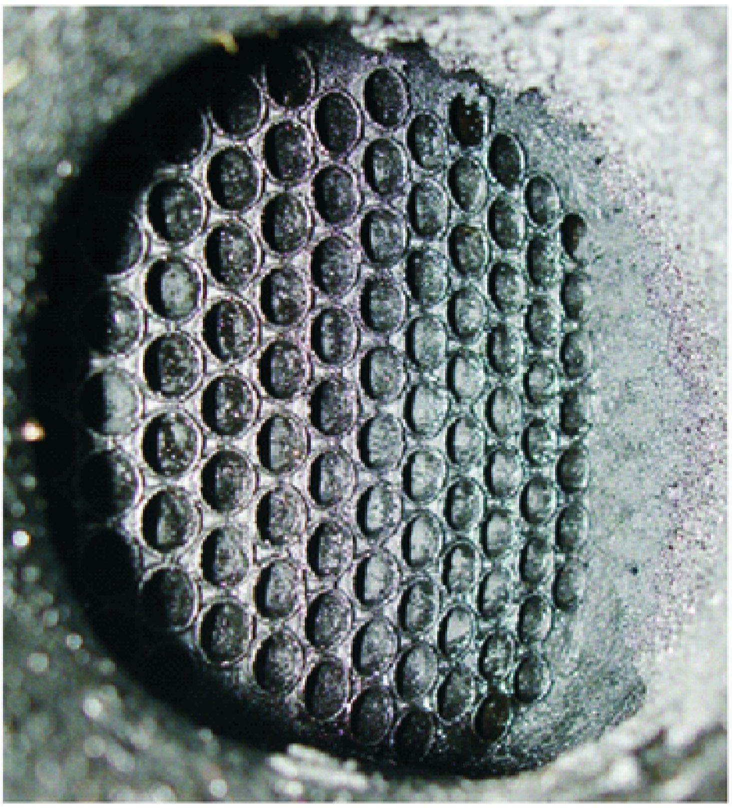
Figure F. Uneven, heavy soot buildup and coolant tracks in the EGR cooler exhaust outlet.

11o00032

Figure G. Very heavy soot buildup in the EGR cooler exhaust outlet.

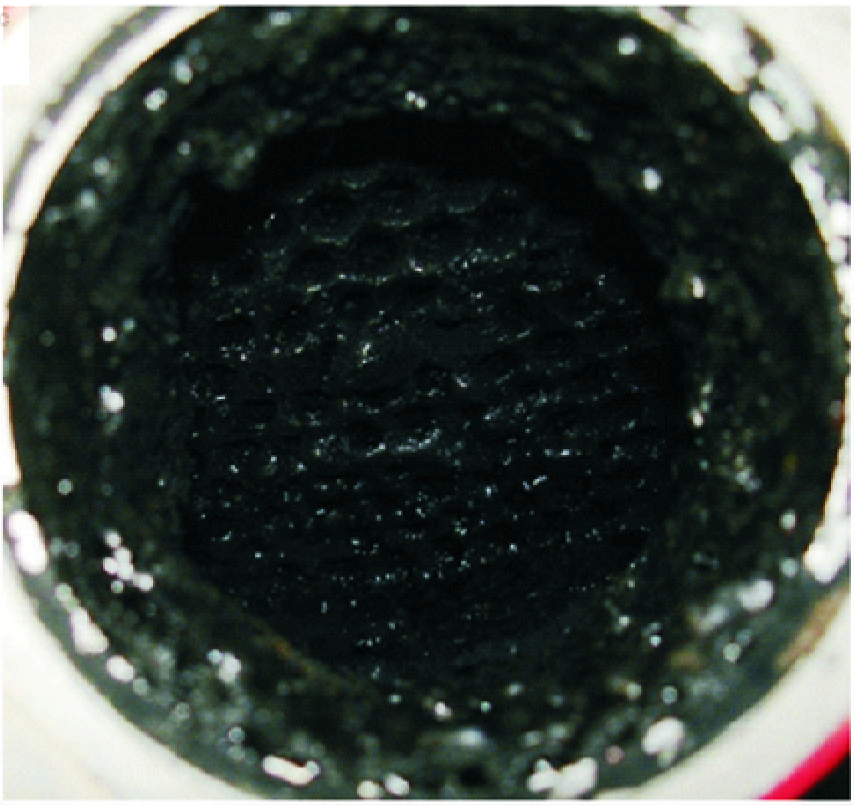
11o00033

Figure H. Very heavy wet soot buildup in the EGR cooler exhaust outlet.



11o00034

Figure I. Strong diesel fuel order and wet soot.



11o00035

Figure J. Sticky or hardened EGR system deposits in the EGR cooler exhaust outlet.

Table 1, Reuse Guidelines for Figures A, B, C, D, E, F, G, H, I, and J		
Figure	**Condition of EGR Cooler Exhaust Outlet**	**Action**
A	Even, light soot	Reuse[1]
B	Even, light wet soot	Reuse[1]
C, D, and E	Uneven soot buildup, coolant tracks	Replace[2]
F	Uneven, heavy soot buildup	Replace[2]
G and H	Very heavy soot buildup	Replace[2]
I	Strong diesel fuel odor and wet soot	Replace
J	Sticky or hardened EGR system deposits	Replace
[1]Reuse **only** if the EGR cooler has passed both the Pressure Decay and Visual Pressure Checks.		
[2]Replace if the EGR cooler has failed the Pressure Decay and/or Visual Pressure checks.		

Note : The X15 CM2350 X116B has a different EGR cooler cross-section as shown in Figure K. The above guidelines apply to this EGR cooler.

Figure K. X15 CM2350 X116B EGR Cooler.

ISB CM2350 and ISL CM2350 have different EGR cooler cross-section than what is as shown in Figure L. Table 1 applies to this EGR cooler.

Figure L. ISB CM2350 and ISL CM2350 EGR Cooler.

Document History

Date	Details
xxxx-xx-xx	Module Created
2012-4-13	Modified to include stationary regen differences specific to Industrial.
2013-3-25	Quick Serve Ticket 86072 requesting clarity on the table at the bottom of Service Bulletin 2883594. The table seems to create confusion because the top of the bulletin says the document is to be used in conjunction with the pressure test but then provides direction on reuse or replacement of the cooler at the bottom. The update adds a note at the bottom of the table for the pictures depicting a failed EGR cooler that says they must fail the pressure decay test and or visual pressure check.
2016-11-7	QSOL Ticket 189582. Added ISX12 CM2350 X102, ISX15 CM2350 X101, X15 CM2350 X114B, X15 CM2350 X116B, and QSX15 CM2350 X106. Added Note and Figure K for X15 CM2350 X116B EGR cooler. Updated to generic title.
2018-3-14	Updated bulletin to reflect generalization of document. Added picture of ISB/ISL EGR cooler.

Date	Details
2019-5-14	Update Metadata

Last Modified: 14-May-2019